

6. Project Testing Phase

FITBUDDY – AI FITNESS PLAN GENERATOR USING GEMINI MODELS

- Unit Testing: Test individual functions, data models, validation, and database operations.
- Integration Testing: Test communication among FastAPI routes, the AI service, database layer, and templates.
- Functional Testing: Verify user input, plan generation, nutrition/recovery output, feedback updates, and administrator views.
- Validation Testing: Check missing inputs, invalid values, empty feedback, unavailable AI responses, and database errors.
- User Interface Testing: Verify navigation, forms, displayed results, readability, and basic responsive behavior.
- Expected Result: The application should accept valid input, generate an appropriate plan, store the result, accept feedback, and return an updated plan while handling errors clearly.